

Multilevel Attention Based U-Shape Graph Neural Network for Point Clouds Learning

Xiaofeng Zou, Kenli Li , and Cen Chen 

Abstract—With the popularity of 3-D sensors in industrial Internet of Things (IIoT), point clouds learning is increasingly important. In this article, we propose a novel multilevel attention based U-shape graph neural network (MAUGNN) for point clouds learning, which can effectively learn the features from low-level to high-level and fuse multiple-level features based on the graph neural networks and attention mechanism. There are three parts in MAUGNN: encoder, decoder, and connections. In the encoder and decoder, we design an attention-based graph convolution to explore the structural information for point clouds. During the encoder, a structure-aware attention pooling is proposed to support down-sampling on point cloud data. To adaptively fuse coarse-grained features from the encoder and fine-grained features from the decoder together, we also propose a structure-aware attention skip connection mechanism. Extensive experiments on popular point cloud datasets demonstrate the superior performance of our MAUGNN over state-of-the-art baselines.

Index Terms—Attention mechanism, encoder-decoder, graph neural networks (GNNs), point clouds.

I. INTRODUCTION

WITH the rapid development of 3-D acquisition technologies, 3-D sensors (e.g., various types of 3-D depth cameras and scanners) are becoming more widespread in industrial Internet of Things (IIoT) [1]. These sensors can conveniently obtain 3-D data with abundant geometric and shape information [2]. Compared with 2-D data (e.g., images), 3-D

data can aid machines to better perceive and understand the surrounding environment, thereby improving the intelligence of robots. Therefore, 3-D perceptions are essential for machines to perform tasks in real environments. At present, 3-D perceptions have numerous applications in various fields of IIoT, including robotics [3], self-driving cars [4], human-computer interaction [5], remote sensing [6], and medical treatment [7].

Point clouds are a popular 3-D data representation that can represent arbitrarily shaped objects in 3-D space. Recently, the analysis of point clouds [8], [9] become an active research topic, which aims to explore the information value of 3-D data. Traditional methods for processing point clouds employ handcrafted features, such as patch context [10] and global point signature [11] to capture the geometric properties of point clouds. However, these approaches are generally designed for specific problems and are difficult to promote to new tasks. Due to the success of convolutional neural networks (CNNs) in various fields [12], some studies [13], [14] have begun to explore deep networks to extract features on point clouds. Since point clouds are sparse and irregular, while classic CNNs require input data with a regular structure, it is not trivial to apply deep networks to point clouds. One popular approach is to first convert disordered point clouds into a regular 3-D voxel network [13]. This approach usually introduces additional processing time and memory and sacrifices the representation accuracy [15].

Some recent work [16]–[19] can directly consume raw point cloud data instead of converting it into multiview voxels. A well-known pioneer work proposed PointNet [16], which utilizes a transformation network to align 3-D shapes, and then applies a global symmetric function to process point clouds. Unfortunately, PointNet handles each point individually, which may cause the loss of local structure message. Following this line, the variant PointNet++ was proposed in [17] to improve the performance by considering local information of neighbor points. However, this approach still treats points independently in the local scale manner and cannot explore the relation between the point and its neighbors (e.g., distance and similarity).

Recent progress in graph neural networks (GNNs) [20]–[22] has extended neural networks to arbitrary structure graphs. GNNs iteratively perform feature aggregation from neighbors by message passing, thus, can discover the complex interactions among instances. A few recent GNN-based methods have shown a superior performance on point clouds tasks [15], [23], [24]. These approaches treat each point in point clouds as a graph node, and generate graph edges based on the neighbors of each point. Then, the point cloud feature matrix and the

Manuscript received May 1, 2020; revised July 21, 2020; September 10, 2020, and November 9, 2020; accepted December 1, 2020. Date of publication December 22, 2020; date of current version September 29, 2021. This work was supported in part by the National Key Research and Development Program of China under Grant 2019YFB2103004, in part by the National Natural Science Foundation of China under Grant 61902120, in part by the National Outstanding Youth Science Program of National Natural Science Foundation of China under Grant 61625202, in part by the International (Regional) Cooperation and Exchange Program of National Natural Science Foundation of China under Grant 61860206011, in part by the Postdoctoral Science Foundation of China under Grant 2019M662768 and Grant 2019TQ0086, and in part by the Natural Science Foundation of Hunan Province under Grant 2020JJ5083. Paper no. TII-20-2194. (Xiaofeng Zou and Cen Chen contributed equally to this work.) (Corresponding author: Kenli Li.)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TII.2020.3046627>.

Digital Object Identifier 10.1109/TII.2020.3046627

generated graph edges are input to the GNNs model to extract features. Nevertheless, the existing GNN-based methods [15], [23], [24] for point clouds have some following limitations: 1) These methods are usually shallow, which are formed by flatly stacking multiple convolutional layers. Thus, they cannot capture adequate hierarchical information. 2) These methods do not consider the attention mechanism when processing point clouds. Existing methods simply assign each point the same weight, which ignores the complex semantic relationship between points.

In this article, we propose a novel multilevel attention based U-shape graph neural network for point clouds learning, shortly termed as MAUGNN. It attempts to exploit multilevel attention mechanisms to explore the structural information of point clouds hierarchically, and uses encoder–decoder architecture to capture local features from multiple spatial levels. Specifically, our proposed MAUGNN contains of the following three parts, i.e., top-down encoder, bottom-up decoder, and skip connections. Both encoder and decoder leverage a new attention-based graph convolutional (AGConv) layer to explore the complex relationships between point clouds for capturing local information. For top-down encoders, we design a graph structure-aware attention pooling (SAPool) to enable down-sampling on point cloud data. The proposed SAPool layer considers both graph structure and node features, and adaptively selects a subset of nodes to achieve graph coarsening. For bottom-up decoder, we design the corresponding graph unpooling (gUnpool) layer to support up-sampling, which restores the down-sampled graph into the original size. Structure-aware attention skip connection (SAConnect) is proposed to adaptively fuse the coarse-grained features learned by the encoder and the fine-grained features learned by the decoder to obtain more hierarchical information.

Our contributions of this article are highlighted as follows.

- 1) To the best of authors' knowledge, it is the first work to design graph encoder–decoder architecture for point clouds learning, which has the advantage of effectively learning the features from low-level to high-level, and capture features of multiple spatial levels.
- 2) We propose a novel graph convolution operation (AGConv), which can better explore the relationship between points and capture local information of point clouds.
- 3) We explore a new attention-based graph pooling method SAPool, which can both considers the graph topology and node features to achieve graph coarsening.
- 4) The proposed SAConnect can reduce the semantic gap between the feature maps of the encoder and decoder.
- 5) Extensive experimental results on popular point cloud datasets show that MAUGNN achieves the best performance compared with state-of-the-art baselines.

II. RELATED WORK

A. Deep Learning on Point Clouds

CNNs have been successfully applied to various challenging artificial intelligence tasks [12], [25]. Inspired by this, many studies [13], [14], [16]–[18] have attempted to apply deep learning to point clouds learning. Some work [13], [14] converted

the sparse and irregular point into a regular 3-D grid data or image collection, and then 3-D convolution can be exploited to handle it. Nevertheless, one great limitation is that the spatial and structural information of the point cloud data may be lost during the transformation process, and incurs additional processing time and computing resources.

On contrary, many research works [16]–[19] directly deal with the original point cloud data by designing appropriate architecture. Qi *et al.* [16] proposed PointNet is a pioneering work in this domain, where multilayer perceptrons (MLPs) are exploited to learn the features of each point, and the max pooling layer is adopted to gather global features from all points. However, each point is handled separately, without considering local structural information. To solve this problem, PointNet++ [17] defined multiscale regions of the input point set and recursively applied PointNet to capture local information, and then aggregated them hierarchically. Motivated by PointNet and PointNet++, Pointwise [18] and PointCNN [19] also emphasized the importance of developing local structural information for 3-D point clouds. However, these models fail to explore the topological relationship between points and their neighbors, which limits the model's ability to learn representations.

B. Graph Neural Networks

Recently, a series of GNNs [20]–[22] have been exploited to process non-Euclidean structure data. GNNs usually follow a neighborhood aggregation algorithm that iteratively updates node representations by aggregating feature informations from neighbors. Therefore, it can discover complex interactions between nodes. Kipf and Welling [20] proposed the graph convolutional network (GCN) that uses average aggregation to generate new node representations. Hamilton *et al.* [21] proposed GraphSAGE, which exploits aggregation schemes such as mean/pooling/LSTM to aggregate features from neighboring nodes. Veličković *et al.* [22] proposed the graph attention network that utilizes attention schemes to determine the weights of neighboring nodes during node aggregation process. Since the encoder–decoder architecture like U-net [26] can perform high-level feature encoding and decoding for network embedding, the latest framework graph U-net [27] developed an encoder–decoder architecture for graphs. However, it can only be applied to graphs with known topology and cannot be directly used for unstructured point cloud.

Some recent work [15], [23], [24] has designed GNN-based methods for point cloud tasks and achieved a superior performance. Since there is no explicit graph structure in unstructured point clouds, these methods treat each point in the point cloud as a graph node, and generate graph edges based on the neighborhood of each nodes. Representation learning is then executed by exploiting GNNs. Specifically, Te *et al.* [23] proposed RGCNN, which dynamically constructs similarity graphs for point clouds through a distance metric function. DGCNN [15] performed k -nearest neighbor algorithm on the feature matrix to construct graph edges, and dynamically updates each layer of the network. Nevertheless, these abovementioned approaches are shallow, which are formed by flatly stacking multiple convolutional

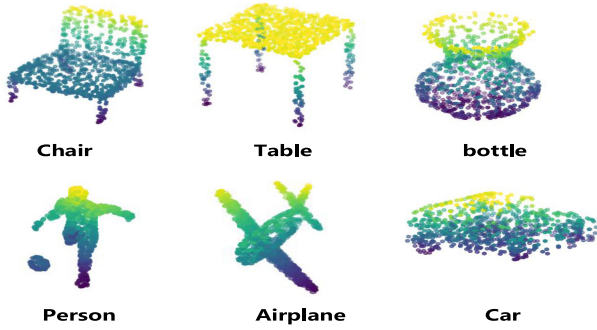


Fig. 1. Illustration of point clouds. Point clouds consist of a set of points, and each point contains several attributes (e.g., coordinates, normal, color).

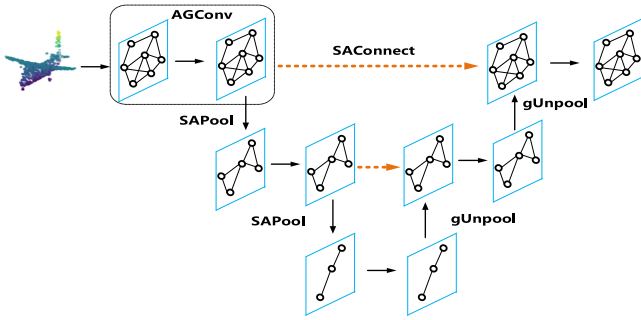


Fig. 2. Overall architecture of the proposed MAUGNN. It is an illustration of a sample of MAUGNN with two blocks in encoder and decoder. Each encoding block contains an AGConv layer and an SAPool layer, and each decoding block contains an AGConv layer and a gUnpool layer. For blocks in the same level, decoding block uses SAConnect to fuse the low-level spatial features from the encoding block.

layers, so that they cannot capture the adequate hierarchical information.

III. METHODOLOGY

A. Problem Formulation

As illustrated in Fig. 1, a point cloud can be represented as a set of 3-D points $\{p_i\}_{i=1}^n$, where $p_i \in \mathbb{R}^m$ is a vector representing the feature of the i th point, such as color and coordinates.

We develop a GNN-based network that directly consumes irregular 3-D point clouds. In our MAUGNN, the input is a feature matrix $X \in \mathbb{R}^{n \times m}$. For the 3-D object classification task that contains q semantic labels, our proposed network outputs the q confidence scores of all candidate classes. For semantic segmentation task, MAUGNN outputs $n \times q$ scores for each of the n points and each of the q labels.

B. Overall Architecture of MAUGNN

The architecture of MAUGNN is illustrated in Fig. 2. MAUGNN takes the raw point cloud with coordinates as input, and follows the typical U-shaped network design, which contains the following three parts: top-down encoder, bottom-up decoder, and skip connection.

The top-down encoder part consists of several encoding blocks, each of which is composed of an AGConv layer

followed by an SAPool layer. The AGConv layer exploits attention mechanism to learn the topological relationship between nodes, and the graph convolution is adopted to aggregate neighborhood messages to capture local information. The SAPool layer considers both graph topology and node features, and reduces the graph size to summarize the higher-level features. In the bottom-up decoder part, we stack the same number of decoding blocks as the encoder part. Each decoding block is composed of an AGConv layer followed by a gUnpool layer. Similar to the encoding block, the AGConv layer captures local information from neighboring nodes, while the gUnpool layer recovers pooled features.

In addition, we propose SAConnect, which is used to connect the encoding blocks and corresponding decoding blocks, which effectively fuses low-level features from the encoder with higher level features from the decoder. Different from the direct addition in the classic U-net design, SAConnect employs the weighted attention mechanism to learn weights, and adaptively fuses low-level and high-level features. The final layer of the MAUGNN is a softmax function mapping the node features to the categories and the cross-entropy meta loss is evaluated for labels of the nodes. The overall process of MAUGNN is shown in Algorithm 1.

The model of MAUGNN is similar to graph U-net [27], which utilizes GCN, gPool, and gUnpool to form a U-shaped neural network for graph representation learning. The differences between MAUGNN and graph U-net are listed as follows. First, AGConv in MAUGNN is different from the GCN layer adopted in graph U-net. AGConv can focus on the most relevant neighbor message during the aggregation phase to dynamically capture local information. Second, SAPool is also different from gPool in graph U-net. During down-sampling, gPool only considers the node features, while SAPool considers both node features and graph structure. Finally, we propose SAConnect to replace the raw skip connection mechanism. The redesigned mechanism can adaptively fuse low-level features from the encoder with higher level features from the decoder, and then further reduce the semantic gap between the feature maps of the encoding block and decoding block.

C. Attention-Based Graph Convolutional (AGConv) Layer

In this section, we describe our proposed AGConv layer in detail. It is well known that point clouds data has some geometric characteristics, but the graph structure is unknown. Therefore, the previous graph convolution layer [20], [22] cannot be directly applied to the point cloud. Our proposed AGConv contains two main processes: graph construction that generates graph edges for unstructured point clouds, and graph convolution that aggregates neighborhood message to generate new node representations.

1) *Graph Construction*: Since an appropriate graph structure has crucial effect on the efficient graph representation learning, it is necessary to select a suitable method for constructing graphs. Previous work [15], [24] utilized k -nearest neighbor algorithm to construct KNN graph, which only considers the locational


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Input: Point cloud  $p_i$ , feature matrix  $X$  and network depth  $depth$ 
Output: Category label  $q$ 
1: Encoder part:
2: for  $l$  in range( $depth$ ) do
3:   Update feature representation  $X^l$  by AGConv
4:   Generate reduced feature matrix by SAPool
                                     # Down-sampling
5: end for
6: Decoder part:
7: for  $l$  in range( $depth$ ) do
8:   Restore the reduced features by gUnpool
                                     # Up-sampling
9:   Adaptively fuse the feature of the encoding block
   and the corresponding decoding block according to
   SAConnect
10:  Update feature representation  $X^l$  by AGConv
11: end for
12: Calculate the category label  $q$  by softmax function
13: return  $q$ 

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After obtaining the selected node indices idx , the new reduced adjacent matrix A^{l+1} and feature matrix X' are calculated through the index operation. The attention scores $\text{attp}_{\text{mask}}^l$ are used to control the information interaction. We adopt the element-wise product of $\text{attp}_{\text{mask}}^l$ and X' to form the output node features X^{l+1} . The calculation process is presented as follows:

$$A^{l+1} = A_{\text{idx}, \text{idx}}^l, X' = X_{\text{idx}}^l, X^{l+1} = X' \odot \text{attp}_{\text{mask}}^l \quad (5)$$

where $\cdot_{\text{idx}}$ denotes the row or column extraction operation based on idx , and $\odot$ is the element-wise product.

E. gUnpool Layer

The up-sampling is the inverse operation of the down-sampling, which is significant in CNN-based U-net. However, due to the particularity of the graph data, previous operations cannot be used directly. To this end, we design a gUnpool layer that can up-sample the graph data. It restores the graph to the original input size while preserves the original structure of the input graph.

During SAPool, we record idx that retains the order information of the selected nodes in the original feature matrix. In the corresponding up-sampling phase, we place the nodes back to the original position in the graph according to the idx to generate the output X^{l+1} . The specific calculation process can be defined as

$$X^{l+1} = F(0_{n \times m}, X^l, \text{idx}) \quad (6)$$

where X^l is the feature matrix, $0_{n \times m}$ is the initially empty feature matrix for the new graph, and $F(0_{n \times m}, X^l, \text{idx})$ is the operation of populating the row vector of X^l to $0_{n \times m}$ feature matrix according to the corresponding node index stored in idx .

F. Structure-Aware Attention Skip Connection

The general U-net network design has a key identity: skip connections, which propagates coarse-grained features from the encoder subnetwork to fine-grained features of the decoder subnetwork, enabling different semantics feature map fusion. The simple addition skip connection is adopted in graph U-net [27]. However, neither the topology of the graph, and the weight between the coarse-grained and the fine-grained features are considered in this approach. Therefore, we design a new skip connection mechanism: SAConnect. The redesigned mechanism can learn the weight between the shallow low-level features and the deep high-level features, and adaptively fuse the two-level features according to the weight, and then further reduce the semantic gap between the feature maps of the encoding block and decoding block.

Fig. 4 shows an example of an SAConnect layer. It has the following three inputs: the feature map of the decoding block $X^l \in \mathbb{R}^{n \times m}$ in l th layer, the feature map of the corresponding encoding block $E^l \in \mathbb{R}^{n \times m}$, and the adjacent matrix $A^l \in \mathbb{R}^{n \times n}$. The output of this layer is the fused feature map, which is the input of next decoding block.

Due to the important role of structural information in graph data, we first inject structural information into the node features. Consistent with Section III-D, we apply the graph convolutions

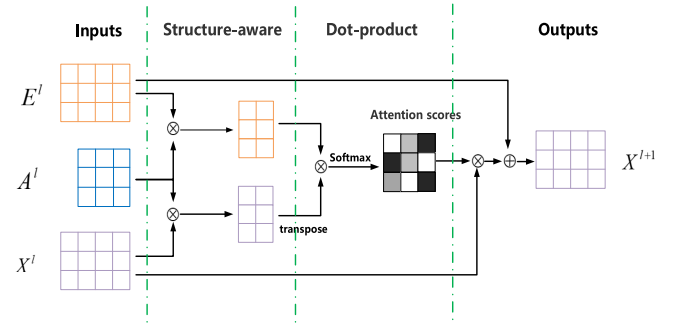


Fig. 4. Illustration of the SAConnect. The SAConnect layer takes the feature map of the decoding block X^l , the feature map of the corresponding encoding block E^l and the adjacent matrix A^l as input, and generates the new node representation X^{l+1} by applying the structure-aware attention.

to X^l and E^l , respectively, to generate new feature representations, i.e.,

$$\begin{aligned} \hat{X}^l &= \tilde{D}^{-\frac{1}{2}} A^l \tilde{D}^{-\frac{1}{2}} X^l \\ \hat{E}^l &= \tilde{D}^{-\frac{1}{2}} A^l \tilde{D}^{-\frac{1}{2}} E^l. \end{aligned} \quad (7)$$

Subsequently, we use self-attention to get the attention coefficients. It is divided into the following two steps: 1) The feature matrices $\hat{X}^l$ and $\hat{E}^l$ are mapped to two feature spaces by using linear transformation layers. 2) We then use dot-product attention as the attention function to calculate the response between the two feature spaces. Note that the commonly used attention functions also include addition attention. Vaswani [28] has proved that dot-product attention is the simpler and more effective method. The attention coefficients e_{ij} is formulated as follows:

$$e_{ij} = \hat{X}_i^l (W_x)^T \cdot W_e \left(\hat{E}_j^l \right)^T \quad (8)$$

where $\cdot$ is the dot-product, the linear transformations $W_x \in \mathbb{R}^{m' \times m}$ and $W_e \in \mathbb{R}^{m' \times m}$ are trainable weight matrices, which are implemented as 1-D convolution layer with filter kernel size of 1. The reason for this setting is to transform the features of nodes into low-dimensional feature space without changing the relative position of nodes, so as to reduce the number of trainable parameters and the computational complexity.

Following the standard setting of attention network, the softmax operator is used to normalize the abovementioned attention coefficients, and get the final attention scores attp_{ij}

$$\text{attp}_{ij} = \text{softmax}(e_{ij}) = \frac{\exp(e_{ij})}{\sum_{k \in N_i} \exp(e_{ik})} \quad (9)$$

where attp_{ij} is the attention degree of j th node of the encoding block to i th node of the decoding block, and vice versa. Since we consider the structural features, we only calculate attp_{ij} for the nodes $j \in N_i$, where N_i represents all neighboring nodes of node i in the graph.

Then, the output of attention connection layer is $X^{l+1} \in \mathbb{R}^{n \times m}$, i.e.,

$$X_i^{l+1} = \sum_{j=1}^N \text{attp}_{ij} W_h X_j^l + E_i^l \quad (10)$$

TABLE I
COMPARATIVE CLASSIFICATION RESULTS ON MODELNET10 AND MODELNET40 BENCHMARKS

Categories	Baseline	Points	ModelNet10 (mAcc)	ModelNet10 (oAcc)	ModelNet40 (mAcc)	ModelNet40 (oAcc)
CNNs	PointNet [16]	1,024	91.2	91.6	86.0	89.2
	PointNet++ [17]	5,000	-	-	-	90.7
	PointCNN [19]	1,024	-	-	88.1	92.2
GNNs	PointGCN [24]	1,024	91.6	91.9	86.1	89.5
	RGCNN [23]	1,024	-	-	87.3	90.5
	DGCNN [15]	1,024	93.0	93.5	90.2	92.2
Ours	MAUGNN	1,024	94.3	94.6	90.7	93.1

The “mAcc” represents mean accuracy (%) and the “oAcc” means overall accuracy (%). The symbol “-” means the results are unavailable. The bold entities indicate the method to obtain the best performance for the different metric.

where the linear transformation $W_h \in \mathbb{R}^{m \times m}$ is learnable weight matrix.

G. Complexity Analysis

Given a graph with n nodes, each with m features. The cost of AGconv is $O(nmm' + \xi mm')$, where ξ is the numbers of edges in the graph and m' is the output features dimensions. The time complexity of SAPool is $O(nm)$, while gUnpool costs $O(n + \xi)$, SAConnect costs $O(nmm')$. We set the number of encoding blocks and decoding blocks to k , the overall complexity is $O(2k(n + \xi)mm' + k(nm + n + \xi + nmm'))$. If we assume that $\xi \gg n$ and $m \approx m'$, the overall complexity is $O(k\xi m^2)$.

IV. EXPERIMENTAL RESULTS

In this section, we compare our proposed MAUGNN with some state-of-the-art models on classification and segmentation tasks. We also visualize experimental results to illustrate the main differences from previous work.

A. Datasets

We evaluate and compare our proposed MAUGNN with state-of-the-art approaches on the following three widely utilized point cloud benchmarks.

- 1) *ModelNet10 and ModelNet40* [30] are two popular 3-D object classification datasets. ModelNet10 consists of 4899 objects from 10 categories, split into 3991 training samples, and 908 test samples. ModelNet40 consists of 12 311 objects belonging to 40 categories, split into 9842 training samples and 2468 test samples. For fair comparison, we adopt the same data preprocessing as [16]. For each object, we evenly sample 1024 points by face area, where each point is represented with 3-D coordinates (x, y, z) .
- 2) *ShapeNet* [30] is popular 3-D object segmentation benchmarks. This dataset consists of 16 681 samples belonging to 16 common categories and 50 parts in total (each category contains 2–6 parts). To facilitate comparison, we follow the train/validation/test split scheme as [15]. For each object, 2048 points with coordinates are uniformly sampled as input.

B. Baselines

We compare our MAUGNN with some representative and state-of-the-art baselines. These models can be categorized into the following two main group: CNN-based and GNN-based models. The classic CNN-based methods include the following: PointNet [16], PointNet++ [17], and PointCNN [19]. Another three GNNs-based methods are also compared as baselines PointGCN [24], RGCNN [23], DGCNN [15]. The abovementioned methods all directly consume raw point cloud data instead of converting it into multiview voxels.

C. Implementation Details

MAUGNN is implemented on PyTorch and run with NVIDIA Tesla P100 GPUs.¹ For fair comparison, we set the same parameter as [15] and [23]. Specifically, The Adam optimize with 0.001 learning rate is utilized to optimize the model. The dimension of node representations is 128, and the batch size is set to 32. In the encoder, we stack three encoding blocks. Each encoding blocks contains a AGConv layer and an SAPool layer. The pooling ratio is set to 0.5. The corresponding decoder is also composed of three decoding blocks, and each decoding block includes a gUnpool layer and an SAPool layer. We use SAConnect between the encoding block and the decoding block. Finally, we utilize the maxpool to obtain a global feature and input it into an MLP layers (512-256-10/40) to obtain the predicted scores. LeakyReLU and Dropout with $p = 0.5$ are adopted to all fully connected layers.

D. Performance Study

1) *Point Cloud Classification*: For classification tasks, we utilize two performance metrics, mean accuracy and overall accuracy, to evaluate performance. Table I and Fig. 5 report the comparison results of our proposed MAUGNN and other baselines on ModelNet10/40 benchmarks. From this table, we have the following observations.

Our proposed MAUGNN consistently outperforms other state-of-the-art baselines on all datasets. On the ModelNet10/40 benchmarks, the overall classification accuracy of MAUGNN (94.6% and 93.1%) is 1.3% and 0.9% higher than DGCNN,

¹The source code as well as the dataset will be provided upon request. It will be released to the public after this article review.

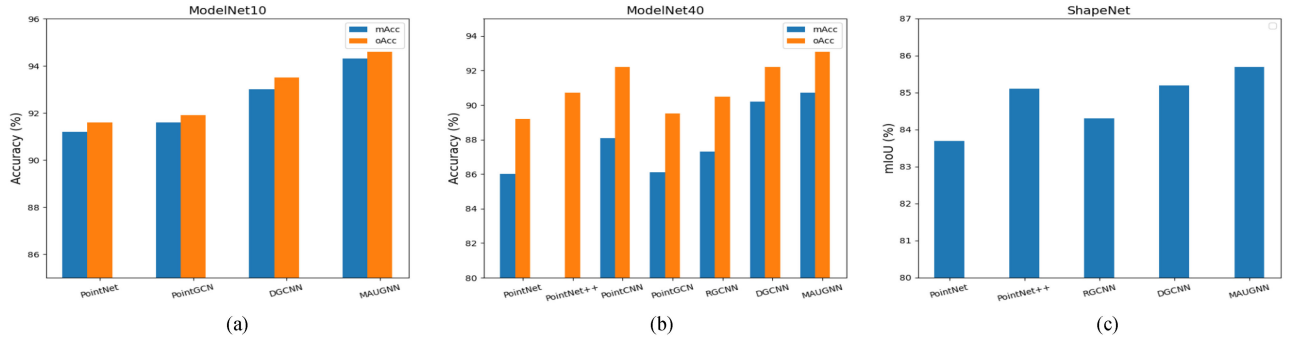


Fig. 5. (a) Results with MAUGNN on ModelNet10. (b) MAUGNN on ModelNet40. The mean accuracy (mAcc) of PointNet++ is unavailable on ModelNet40. (c) MAUGNN on ShapeNet.

TABLE II
COMPARATIVE SEGMENTATION RESULTS ON SHAPENET BENCHMARK

Categories	Baseline	mIoU	aero	bag	cap	car	chair	earphone	guitar	knife	lamp	laptop	motor	mug	pistol	rocket	skateboard	table
CNNs	PointNet [16]	83.7	83.4	78.7	82.5	74.9	89.6	73.0	91.5	85.9	80.8	95.3	65.2	93.0	81.2	57.9	72.8	80.6
	PointNet++ [17]	85.1	82.4	79.0	87.7	77.3	90.8	71.8	91.0	85.9	83.7	95.3	71.6	94.1	81.3	58.7	76.4	82.6
GNNs	RGCNN [23]	84.3	80.2	82.8	92.6	75.3	89.2	73.7	91.3	88.4	83.3	96.0	63.9	95.7	60.9	44.6	72.9	80.4
	DGCNN [15]	85.2	84.0	83.4	86.7	77.8	90.6	74.7	91.2	87.5	82.8	95.7	66.3	94.9	81.1	63.5	74.5	82.6
Ours	MAUGNN	85.7	84.1	82.9	84.9	79.4	90.8	76.5	90.8	88.2	83.4	96.5	75.3	96.0	82.7	62.2	76.2	82.2

The bold entities indicate the method to obtain the best performance for the different metric.

respectively. It demonstrates that our proposed MAUGNN can effectively capture local features of multiple spatial levels while encoding and decoding high-level features.

The performance of GNN-based methods (e.g., PointGCN, RGCNN, DGCNN) is generally superior to CNN-based methods (e.g., PointNet, PointNet++), which effectively validates the superiority of GNNs for processing irregular data like point clouds. Compared with the CNN-based methods, it can fully extract local information of neighboring points to improve the network performance. Our proposed MAUGNN is superior to all GNN-based methods, which verifies the effectiveness of MAUGNN considering the hierarchical attention mechanism and encoder-decoder structure.

2) *Point Cloud Segmentation*: For segmentation tasks, we utilize the Intersection-over-Union (IoU) to evaluate performance of the networks. The mIoU is the average of IoU for all test categories. Table II and Fig. 5(c) summarize the comparison results on ShapeNet benchmark. We can observe that the proposed MAUGNN achieves the highest performance scores. Meanwhile, MAUGNN is superior to other baseline methods in most categories. It demonstrates that MAUGNN can effectively learn the local information of the point clouds by utilizing the multilevel attention mechanism, and exploit the encoder-decoder framework to learn the feature representations from low-level to high-level, and capture features of multiple spatial levels. More detailed segmentation results of MAUGNN are demonstrated in Fig. 6. Obviously, our segmentation results are visually closer to the ground truth than the baseline DGCNN.

E. Ablation Studies and Visualization

1) *Effectiveness of the Encoder-Decoder Architecture*: To evaluate the effectiveness of the encoder-decoder architecture, we explore an variant: MAUGNN_{NED} (without encoder-decoder) means removing the encoder-decoder architecture

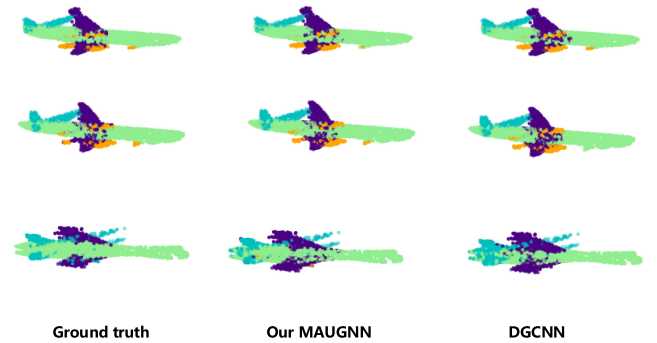


Fig. 6. Visual comparison of segmentation results. From left to right: ground truth, our MAUGNN and DGCNN.

TABLE III
ABLATION STUDIES OF ENCODER-DECODER ARCHITECTURE

Baseline	ModelNet10 (mAcc/oAcc)	ModelNet40 (mAcc/oAcc)	ShapeNet (mIoU)
PointNet [16]	91.2/91.6	86.0/89.2	83.7
DGCNN [15]	93.0/93.5	90.2/92.2	85.2
MAUGNN _{NED}	93.3/93.9	90.3/92.7	85.4
MAUGNN	94.3/94.6	90.7/93.1	85.7

The bold entities indicate the method to obtain the best performance for the different metric.

and adopting the same architecture as PointNet [16] and DGCNN [15]. Table III shows the experimental results of MAUGNN and its variants on all benchmarks. We can observe that MAUGNN outperforms MAUGNN_{NED} in all evaluation metrics, which effectively proves that the encoder-decoder can extract the local features of multiple spatial levels hierarchically. After removing the encoder-decoder architecture, MAUGNN_{NED} also achieves the better performance than PointNet and DGCNN.

TABLE IV
ABLATION STUDIES OF THE MULTILEVEL ATTENTION MECHANISM

Baseline	ModelNet10 (mAcc/oAcc)	ModelNet40 (mAcc/oAcc)	ShapeNet (mIoU)
MAUGNN _{NAGC}	93.5/94.0	90.3/92.6	85.4
MAUGNN _{gPool}	93.8/94.2	90.4/92.9	85.5
MAUGNN _{NC}	93.5/93.9	90.5/92.5	85.4
MAUGNN _{Add}	94.0/94.2	90.6/92.9	85.5
MAUGNN	94.3/94.6	90.7/93.1	85.7

The bold entities indicate the method to obtain the best performance for the different metric.

TABLE V
COMPLEXITY COMPARISON ON MODELNET10 BENCHMARK

Baseline	Model size(MB)	Time(ms)	oACC(%)
PointNet [16]	5.62	16.6	91.6
DGCNN [15]	3.40	26.3	93.5
MAUGNN	3.43	23.8	94.6

The bold entities indicate the method to obtain the best performance for the different metric.

2) Effectiveness of the Multilevel Attention Mechanism: Some variants of MAUGNN are explore to evaluate the effectiveness of multilevel mechanism.

- 1) MAUGNN_{NAGC} (without AGConv) denotes removing the proposed AGConv and adopting the graph convolution based on KNN graphs [24].
- 2) MAUGNN_{gPool} denotes utilizing gpool proposed in [27] instead of SAPool.
- 3) MAUGNN_{NC} (without SAConnect) means that removes the skip-connection.
- 4) MAUGNN_{Add} denotes that the attention mechanism is not applied to skip-connection, but directly uses simple addition operations.

Table IV provides the comparison results of MAUGNN and all its variants on all benchmarks. The results show that MAUGNN is superior to MAUGNN_{NAGC}. It indicates that AGConv can effectively explore the relationship between points and capture local information of the point cloud. Compared with MAUGNN_{gPool}, MAUGNN adopts SAPool to consider node features and structural information while down-sampling, which enables the model to summarize higher level features. We also can observe that MAUGNN and MAUGNN_{Add} perform better than MAUGNN_{NC}, which effectively proves the effectiveness of skip connection. Furthermore, the attention mechanism is exploited in skip connection that can adaptively combine coarse-grained and fine-grained features together for better performance. Overall, multilevel attention can ensure that the model selectively focuses on the most relevant parts of the task and greatly improve the learning ability.

3) Model Complexity Analysis: We utilize the classification experiment on ModelNet10 to evaluate the model complexity. **Table V** shows that our proposed MAUGNN has achieved the best tradeoff between the computational complexity, model complexity (model size), and classification accuracy.

4) Visualization: **Fig. 7** shows the t-SNE visualization results of our MAUGNN and PointGCN on ModelNet10. Compared to PointGCN, MAUGNN shows better clustering on point clouds,

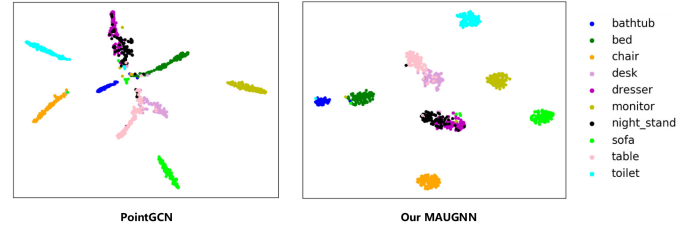


Fig. 7. T-SNE visualization on the ModelNet10 for PointGCN and our MAUGNN. Each color represents a category.

TABLE VI
INFLUENCE OF THE NETWORK DEPTHS ON MODELNET40 BENCHMARK

Network depths	2	3	4	5
ModelNet40 (mAcc)	90.0	90.7	90.7	90.5
ModelNet40 (oAcc)	92.2	93.1	92.9	92.8

Bold entities indicates that the accuracy of MAUGNN on all datasets is superior to other baselines.

and the boundaries between categories are more obvious. It indicates that MAUGNN can learn more meaningful representations by encoding and decoding high-level features. In addition, we discover that there are the following two groups of categories that are difficult to classify correctly: 1) nightstand and dresser and 2) table and desk. This is because these categories share similar 3-D structures, and our MAUGNN captures similar structural relationships for these objects, thereby affecting subsequent feature learning and classification.

F. Network Depth Study

Since the network depth represented by the number of encoding and decoding blocks is an important hyper-parameter in MAUGNN, we evaluate the impact of different network depths on MAUGNN performance on ModelNet40. The results are summarized in **Table VI**. We can discover that the performance improves as network goes deeper until it reaches 3. After that, with the increase of network depth, it will lead to overfitting problem and more interference.

V. CONCLUSION

3-D perceptions are eagerly in demand in IIoT, due to it can aid machines to perceive and understand the environment. In this article, we proposed a novel MAUGNN that directly process irregular 3-D point clouds. It combined attention mechanism with U-type network to achieve better representation learning capabilities. MAUGNN exploited the multilevel attention mechanism to learn the relationship between points in point clouds for capturing local information. The encoder-decoder architecture can ensure that the model can extract features from multiple spatial levels. Based on that, we further design the operations of AGConv, SAPool, and SAConnect. Extensive experiments on the popular point cloud datasets have demonstrated that MAUGNN outperforms existing challenging baselines. For instance, in the ModelNet40 object classification task, our model achieved a state-of-the-art accuracy of 90.7%. In the ShapeNet object segmentation task, mIoU reached 85.7%. For future research,

we plan to explore more powerful frameworks to distinguish the objects with similar structures. Another promising direction is to investigate the impact of other features such as normal.

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